Near-uniform sampling of factored smooth integers in expected polynomial time

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Abstract

Given integers $2 \le y \le X$, we consider the generation of an integer chosen nearly uniformly from the y-smooth integers at most X, together with its complete prime factorization. We give an unconditional algorithm using fair random bits whose output law is within total variation 2^{-h} of uniform and whose expected bit complexity is bounded by a fixed polynomial in $\lceil \log_2 X \rceil + h$, uniformly in y. The algorithm requires no prime list, smooth-number counting oracle, saddle-point evaluation or general integer-factorization routine. Its construction combines tilted independent geometric prime exponents with sparse proposals over dyadic integer intervals. An explicitly filtered grid contains a useful tilt without computing it. Uniform smooth-number estimates give polynomial acceptance, and certified finite-precision primitives preserve both the distributional guarantee and the expected work bound. The guarantee concerns expected bit complexity; no practical running-time comparison is asserted.

Keywords: smooth integers; random sampling; bit complexity; rejection sampling; prime factorization.

1 The sampling problem

An integer is y-smooth if each of its prime factors is at most y. Sampling such an integer together with its factorization requires control of two different aspects of the output: the integer must have the prescribed distribution, and its factors must be available without factoring an arbitrary integer after it has been chosen. For unrestricted integers, the algorithms of Bach [1] and Kalai [2] achieve these objectives by constructing the factorization during generation. A cutoff on the permitted primes changes the acceptance probabilities, however, and a procedure that is efficient without that cutoff can become exponentially slow in the binary input length.

Write

$$\mathcal{S}(X, y) = \{n \in \mathbb{Z} : 1 \le n \le X,\ P^+(n) \le y\}, \qquad \Psi(X, y) = |\mathcal{S}(X, y)|,$$

where $P^+(n)$ is the largest prime factor of $n > 1$ and $P^+(1) = 1$. Let $U_{X,y}$ be the uniform probability measure on this set. Throughout the paper, X and y are integers with $2 \le y \le X$, all logarithms without a subscript are natural, and

$$B = \lceil \log_2 X \rceil.$$

For probability measures on a finite set, we use $d_{\mathrm{TV}}(\mu, \nu) = \frac{1}{2} \sum_n |\mu(n) - \nu(n)|$.

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Theorem 1.1. There is an unconditional randomized algorithm which, given binary integers $2 \le y \le X$ and an integer $h \ge 1$, uses fair random bits, terminates almost surely, and returns $n \in \mathcal{S}(X, y)$ together with its correct complete prime factorization. Its output law $\mu_{X,y,h}$ satisfies

$$d_{\mathrm{TV}}(\mu_{X,y,h}, U_{X,y}) \le 2^{-h}. \tag{1.1}$$

There is a fixed polynomial P, independent of X, y, h, such that its expected number of bit operations is at most $P(B + h)$. The algorithm requires no precomputed list of primes, exhaustive enumeration as preprocessing, evaluation of $\Psi(X, y)$, or oracle for a prime-counting function, an Euler product, a saddle point or integer factorization.

The accuracy cost is polynomial in $h = \log_2(1/2^{-h})$, rather than in the binary encoding length of h. The bound is uniform over all cutoffs, including $y = 2$ and $y = X$. Every accepted output and its factorization are correct; approximation affects only the output probabilities. A rare execution may inspect many integer positions, so the theorem makes an expected, rather than a worst-case, running-time assertion.

Bach and Sorenson [3] study the same factored smooth-integer generation task through enumeration and unranking. Their January 2026 revision includes an exact enumeration method and faster variants whose proved guarantees trade cutoff dependence, additional hypotheses and an asymptotic rank criterion. In particular, Theorem 5.1 retains roughly linear dependence on the cutoff through prime enumeration; Theorems 5.2 and 5.3 use additional assumptions. Those statements do not provide the combination in Theorem 1.1 of unconditional expected bit complexity polynomial in the input length, uniformity in the cutoff, and a prescribed total-variation error. An asymptotically small relative rank error alone does not imply a prescribed total-variation bound: mapping each uniform rank to a neighboring even rank can leave relative rank errors small while concentrating the distribution on half the ranks. This example concerns the implication between criteria, not the distribution of their implementation.

The product distributions used below are established sampling constructions. Kalai's method supplies the unrestricted geometric law and rejection correction [2], and Dirichlet samplers organize multiplicative structures through weights of the form n^{-s} [4]. Sparse arithmetic proposals and finite-bit analysis also have precedents in Bach's construction [1]. The additional work here is to choose and simulate a useful tilt uniformly over the cutoff, then control finite-precision errors after rejection. The analysis uses the classical smooth-number estimates of Hildebrand and Tenenbaum [5], in the formulation of their later survey [6], while exact primality decisions use the unconditional polynomial-time theorem of Agrawal, Kayal and Saxena [7].

Smooth integers also occur in factorization and discrete-logarithm algorithms used in cryptanalysis [8]. The present result concerns their near-uniform generation together with their complete factorizations and does not establish a cryptanalytic speedup.

We first derive the rejection law and show how to generate its active prime factors without listing the primes. We then construct a finite parameter grid and establish a uniform acceptance bound. The last part replaces the ideal real-probability decisions by bounded fair-bit routines and proves that the resulting distribution and expected work satisfy Theorem 1.1.

2 A tilted product law and sparse proposals

For $s > 0$, set

$$Z_y(s) = \prod_{p \le y} (1 - p^{-s})^{-1}.$$

This Euler product is an object of the analysis; the algorithm will not evaluate it. Give each prime $p \le y$ an independent exponent A_p with

$$\Pr(A_p = k) = (1 - p^{-s})p^{-sk}, \qquad k = 0, 1, 2, \ldots.$$

The integer $N = \prod_{p \le y} p^{A_p}$ then has probability

$$Q_s(n) = \frac{n^{-s}}{Z_y(s)} \tag{2.1}$$

at every positive y-smooth integer n. Reject $N > X$; otherwise accept with probability $(N/X)^s$. For each $n \in \mathcal{S}(X, y)$, the probability of generating and accepting n in one trial is

$$Q_s(n)(n/X)^s = \frac{1}{X^s Z_y(s)}. \tag{2.2}$$

Thus the conditional output is exactly uniform, and the acceptance probability is

$$a_s = \frac{\Psi(X, y)}{X^s Z_y(s)}. \tag{2.3}$$

Choosing s from any finite mixture preserves this uniform conditional law: each component gives the same accepted mass to every member of $\mathcal{S}(X, y)$, although the total acceptance probability depends on the component.

The choice $s = 1$ recovers the relevant geometric mechanism in Kalai's algorithm. It does not suffice uniformly in y. When $y = 2$, the permitted integers form the set $\{2^k : 0 \le k \le \lfloor \log_2 X \rfloor\}$, and

$$a_1 = \frac{\lfloor \log_2 X \rfloor + 1}{2X}. \tag{2.4}$$

The expected number of trials is then exponential in B. Allowing the parameter to vary will repair this acceptance cost, but first we must generate the law in (2.1) without processing every prime.

For $b = 1, \ldots, \lfloor \log_2 y \rfloor$, use the integer block

$$I_b = \{2^b, \ldots, \min(2^{b+1} - 1, y)\}, \qquad r_b = 2^{-bs}.$$

Generate independent Bernoulli positions of probability r_b in this block by geometric skipping. At a proposed integer m, toss a further coin of probability $(2^b/m)^s$ and retain m only if this coin succeeds and m is prime. Each prime $p \in I_b$ is retained with probability

$$r_b(2^b/p)^s = p^{-s},$$

independently of all other primes. Conditional on retaining p, give it exponent $1 + G_p$, where $\Pr(G_p = k) = (1 - p^{-s})p^{-sk}$. The resulting exponent has exactly the law of A_p: the zero value has probability $1 - p^{-s}$, and a positive value k has probability $(1 - p^{-s})p^{-sk}$.

The expected number Λ_s of proposed integer positions, counting composites and ignoring early rejection, obeys

$$\Lambda_s = \sum_b |I_b| 2^{-bs} \le \sum_b 2^{b(1-s)} \le B \max(1, y^{1-s}). \tag{2.5}$$

Geometric skipping therefore avoids a cost proportional to the block length when the proposal probability is small. An exact deterministic polynomial-time primality test [7] suffices because every proposed integer has at most $B + 1$ bits.

There is no need to store an exponent that would make the product exceed X. If the current product is n, let $e_{\max}$ be the largest nonnegative integer satisfying $np^{e_{\max}} \le X$. It can be found using integer multiplication and comparison. If $e_{\max} = 0$, reject the trial. Otherwise generate $\min(G_p, e_{\max})$; its value $e_{\max}$ is an overflow outcome and causes rejection. Any smaller value gives the exponent $1 + G_p$ and an updated product at most X. Early rejection removes exactly paths that cannot lead to a permitted output, because all later factors are at least one. All stored integers in these operations have $O(B)$ bits; multiplication can stop as soon as the bound X is exceeded.

3 A computable grid of parameters

We require a list containing a useful tilt while excluding choices whose sparse proposals would be too expensive. The following elementary estimate fixes a numerical constant in that list.

Lemma 3.1. For $x \ge 2$,

$$\theta(x) := \sum_{p \le x} \log p \ge \frac{x}{2048}. \tag{3.1}$$

Proof. Write $\psi(x) = \sum_{p^a \le x} \log p$. Each prime valuation of $\binom{2n}{n}$ is at most $\lfloor \log_p(2n) \rfloor$, so this binomial coefficient divides $\operatorname{lcm}(1, \dots, 2n)$. Since it is the largest of the $2n+1$ binomial coefficients whose sum is 4^n,

$$\psi(2n) \ge 2n \log 2 - \log(2n+1).$$

Taking $n = \lfloor x/2 \rfloor$ and removing higher prime powers gives

$$\theta(x) \ge (x-2)\log 2 - \log(x+1) - \sqrt{x}\log x. \tag{3.2}$$

Indeed, only primes at most $\sqrt{x}$ contribute to $\psi(x) - \theta(x)$, and each contributes at most $\log x$. After division by x, the right side of (3.2) exceeds $1/4$ at $x = 1024$ and increases thereafter: both $[2\log 2 + \log(x+1)]/x$ and $(\log x)/\sqrt{x}$ decrease on this range. For $2 \le x < 1024$, the estimate $\theta(x) \ge \log 2 > x/2048$ completes the proof. $\square$

Define the nonempty integer set

$$J = \{B, B+1, \dots, 4B\} \ \cup \ \{1 \le j < B : y^{B-j} \le (2048B)^B\}, \qquad s_j = \frac{j}{B}. \tag{3.3}$$

The comparisons in (3.3) involve integers of $O(B^2 + B\log B)$ bits and can be performed in polynomial time. Each retained parameter satisfies

$$\Lambda_{s_j} \le 2048B^2. \tag{3.4}$$

To prove that one of these parameters also has adequate acceptance, introduce $L = \log X$ and the unique number $\alpha > 0$ satisfying

$$\sum_{p \le y} \frac{\log p}{p^\alpha - 1} = L. \tag{3.5}$$

The left side is $\mathbb{E}_{Q_s} \log N$ at $s = \alpha$, and it decreases continuously from infinity to zero as s increases. Thus the saddle point centers the logarithmic size of the proposal at the upper bound $\log X$. It is used only to establish the quality of the grid; it is not computed.

We have $\alpha < 4$. In fact,

$$\sum_{p \le y} \frac{\log p}{p^4 - 1} \le \frac{16}{15} \sum_{n \ge 2} \frac{\log n}{n^4} \le \frac{8}{15} \sum_{n \ge 2} n^{-3} \le \frac{2}{15} < \log 2 \le L, \tag{3.6}$$

where $\log n \le n/2$ and $\sum_{n\ge 2} n^{-3} \le 1/4$. Lemma 3.1 and (3.5) also give

$$L \ge \frac{\theta(y)}{y^\alpha - 1} \ge \frac{y^{1-\alpha}}{2048}. \tag{3.7}$$

Since $L \le B$, the upward grid point

$$s_* = \frac{\lceil B\alpha \rceil}{B}$$

belongs to (3.3). If $s_* < 1$, equation (3.7) ensures that its filter inequality holds; if $s_* \ge 1$, it is retained automatically. Furthermore, $Z_y(s)$ decreases with s, so

$$X^{s_*} Z_y(s_*) \le X^{1/B} X^\alpha Z_y(\alpha) \le 2X^\alpha Z_y(\alpha).$$

Consequently

$$a_{s_*} \ge \frac{1}{2} a_\alpha. \tag{3.8}$$

The grid contains a parameter within this factor of the saddle-point acceptance while imposing a polynomial proposal cost on every parameter. It also contains $s = 1$, which will handle a complementary range.

4 Acceptance uniformly over the cutoff

Choose j uniformly from J at the start of each trial, independently of all subsequent choices, and let a be the resulting ideal acceptance probability. We establish two lower bounds for a. A polynomial bound controls the expected number of restarts; an explicit, weaker bound will determine the finite precision without requiring a numerical value for an analytic asymptotic constant.

Proposition 4.1. There is an absolute constant $c_0 > 0$ such that

$$a \ge \frac{c_0}{B^2} \tag{4.1}$$

for every $2 \le y \le X$.

Proof. Put $u = L/\log y$ and

$$\phi_2(\alpha, y) = \sum_{p \le y} \frac{(\log p)^2 p^\alpha}{(p^\alpha - 1)^2}.$$

Under the product law Q_α, this sum is $\operatorname{Var}(\log N)$, since the prime exponents are independent. The saddle-point estimate below relates the accepted mass near the size bound to this logarithmic spread.

The saddle-point estimate of Hildebrand and Tenenbaum [5, Theorem 1; 6, Theorem 1.6, equations (1.22)–(1.25)] states, uniformly for $X \ge y \ge 2$, that

$$a_\alpha = \frac{1 + O(1/u + (\log y)/y)}{\alpha\sqrt{2\pi\phi_2(\alpha, y)}}, \tag{4.2}$$

with an absolute implied constant. Choose fixed thresholds U, Y large enough that the absolute value of the relative error in (4.2) is at most $1/2$ when $u \ge U$ and $y \ge Y$. The thresholds enter only the analysis.

For $t \ge 0$, $t/(1 - e^{-t}) \le 1 + t$. Applying this with $t = \alpha \log p$ and then using (3.5) yields

$$\begin{aligned}\alpha^2\phi_2(\alpha,y) &= \sum_{p\le y}\frac{\alpha\log p}{p^\alpha-1}\frac{\alpha\log p}{1-p^{-\alpha}}\\ &\le (1+\alpha\log y)\alpha L \le 20B^2.\end{aligned} \tag{4.3}$$

Thus (4.2) gives $a_\alpha \ge c/B$ in the range $u \ge U$, $y \ge Y$, and (3.8) gives the same order at s_*.

If $1 \le u \le U$, the classical Dickman estimate on fixed compact u-intervals [6, equation (1.8) and Theorem 1.1] gives $\Psi(X,y) \ge c_U X$ for a positive constant depending only on the fixed U. For sufficiently large y, the uniform error is small and the Dickman function satisfies $\rho(u) \ge \rho(U) > 0$. When y is below this fixed threshold, $X \le y^U$ leaves only finitely many integer input pairs, whose positive density ratios can be absorbed into c_U. The Euler-product estimate in [6, display between (1.3) and (1.4)] bounds $Z_y(\sigma) \le C\log y$ for $\sigma = 1 - 1/(2\log y) < 1$. Monotonicity gives $Z_y(1) \le Z_y(\sigma) \le C\log y$. Hence

$$a_1 = \frac{\Psi(X,y)}{XZ_y(1)} \ge \frac{c'}{B}. \tag{4.4}$$

It remains to control bounded y, where the relative error in (4.2) need not be small. Set $m = \pi(y)$, $w_p = \log p$ and $z = \alpha L$. The inequalities

$$\frac{1}{t} - 1 \le \frac{1}{e^t-1} \le \frac{1}{t}$$

and (3.5) imply

$$\frac{m}{m+1} \le z \le m. \tag{4.5}$$

For the lower bound, $L \ge m/\alpha - \sum_{p\le y}\log p$, and $\sum_{p\le y}\log p \le mL$. The exponent vectors counted by $\Psi(X,y)$ cover the nonnegative simplex $\sum_p t_p w_p \le L$ by their unit cubes: coordinatewise flooring takes each point of the simplex to a counted vector. Comparing volumes therefore gives

$$\Psi(X,y) \ge \frac{L^m}{m!\prod_{p\le y} w_p}. \tag{4.6}$$

Since $(1-e^{-t})^{-1} \le 1 + 1/t$, equations (2.3), (4.5) and (4.6) imply

$$\begin{aligned}a_\alpha &\ge \frac{z^m}{m!e^z\prod_{p\le y}(1+zw_p/L)}\\ &\ge \frac{[m/(m+1)]^m}{m!e^m(1+m)^m} > 0.\end{aligned} \tag{4.7}$$

For $y \le Y$, only finitely many values of m occur, so this supplies an absolute positive lower bound. The three ranges cover all inputs. At least one retained parameter consequently has acceptance at least c/B, after reducing c if necessary. Finally, $|J| \le 4B$, and averaging proves (4.1). □

Lemma 4.2. Define

$$H_0 = 32768B^3. \tag{4.8}$$

Every retained parameter satisfies $a_{s_j} \ge 2^{-H_0}$, and so does their mixture.

Proof. Every $s_j \ge 1/B$. Monotonicity in s and $1 - e^{-t} \ge t/2$ for $0 \le t \le \log 2$ give

$$\frac{1}{1-2^{-s_j}} \le \frac{1}{1-2^{-1/B}} \le 4B.$$

Moreover, the expected number of activated primes is at most the expected number of proposed positions. By (3.4),

$$\begin{aligned}\log Z_y(s_j) &\le \sum_{p\le y} \frac{p^{-s_j}}{1-p^{-s_j}} \\ &\le 4B \sum_{p\le y} p^{-s_j} \le 8192B^3.\end{aligned}$$

As $s_j \le 4$ and $L \le B$, we have $\log[X^{s_j} Z_y(s_j)] \le 8196B^3$. The integer 1 belongs to $\mathcal{S}(X,y)$, and $\log 2 > 1/2$, so (2.3) gives $a_{s_j} \ge [X^{s_j} Z_y(s_j)]^{-1} \ge 2^{-H_0}$. □

5 Finite-bit probability decisions

The ideal construction uses real-valued probabilities, whereas Theorem 1.1 permits only fair random bits. We now give bounded routines for those decisions. A bound on every possible trial length will let us control the effects of numerical error even on rare, unusually long executions.

5.1 Certified rational powers

The base probabilities used for positional skipping, thinning, prime exponents and final acceptance have the form

$$v = (a/b)^{j/B}, \qquad 1 \le a \le b \le X, \quad 1 \le j \le 4B, \tag{5.1}$$

or their complements. Values 0 and 1 are handled exactly. For an integer $R \ge 1$, comparison of a dyadic number $d/2^R$ with v is equivalent to comparison of the integers

$$d^B b^j \quad \text{and} \quad 2^{RB} a^j. \tag{5.2}$$

Binary search among the dyadic values therefore produces a certified enclosure of width 2^{-R} in deterministic time polynomial in $B+R$. The powers d^B, b^j and a^j have exponents at most $4B$; the factor 2RB is constructed by a binary shift. All integers in (5.2) have polynomially many bits.

We will also need q^ℓ for $0 \le \ell \le X$, where q is a number in (5.1) or its complement. For $\ell = 0$ return 1. Otherwise approximate q to $R = P + 3B + 10$ bits and use left-to-right binary exponentiation, initialized at the leading binary digit, with fixed-point multiplication at the same precision and values clamped to $[0,1]$. There are at most $2B$ multiplications. The input error is amplified by at most $\ell \le 2^B$; each rounding error can likewise be amplified by at most 2^B through the remaining products. The total error is at most

$$(2B+2)2^B 2^{-R} < 2^{-P}. \tag{5.3}$$

For instance, this follows by viewing the repeated-squaring computation as a product computation in $[0,1]$: a local perturbation contributes at most its multiplicity in the final product, which is bounded by ℓ. Certified intervals, or fixed-point values with this propagated error allowance, give the claimed enclosure. Thus even an exponent ℓ exponential in B requires only polynomial bit work in $B+P$; no exact rational denominator of length proportional to ℓ is constructed.

5.2 A truncated geometric routine

For $q \in [0,1]$ and $N \geq 1$, define $K(q,N)$ by

$$\Pr(K=k) = (1-q)q^k \quad (0 \leq k < N), \qquad \Pr(K=N) = q^N. \tag{5.4}$$

This includes the endpoint laws $K(0,N) = 0$ and $K(1,N) = N$. For $q < 1$ it is the number of failures before a success, truncated at N. Its distribution function is

$$F(k) = 1 - q^{k+1} \quad (0 \leq k < N), \qquad F(N) = 1.$$

Lemma 5.1. If $N \leq X$ and q has a form covered by Section 5.1, a bounded fair-bit routine produces a law within total variation 2^{-t} of $K(q,N)$ in deterministic time polynomial in $B+t$.

Proof. Handle $q = 0, 1$ exactly. Otherwise put $P = B + t + 4$ and draw P fair bits to obtain a uniform dyadic number $U_D \in \{0, 2^{-P}, \ldots, 1-2^{-P}\}$. Start with low $= 0$, high $= N$. While low $<$ high, set mid $= \lfloor (\text{low} + \text{high})/2 \rfloor$ and compute an approximation $\widehat{F}(\text{mid})$ with absolute error at most 2^{-P}. If $\widehat{F}(\text{mid}) > U_D$, replace high by mid; otherwise replace low by mid $+ 1$. Return low. This always terminates in $O(B)$ comparisons with an output in $\{0, \ldots, N\}$.

Couple U_D to a continuous uniform variable U by $U_D = 2^{-P}\lfloor 2^P U \rfloor$. Outside the union of intervals of radius $2 \cdot 2^{-P}$ around the N true thresholds $F(0), \ldots, F(N-1)$, every possible comparison agrees with comparison of U against the corresponding exact threshold. The numerical and exact binary searches then follow the same path and return the same value. The exceptional event has probability at most

$$4N2^{-P} \leq 2^{-t}/4 \leq 2^{-t}. \tag{5.5}$$

This coupling proves the total-variation bound. The approximated thresholds need not be monotone: outside the exceptional event their queried comparisons agree with the exact monotone distribution function, and on the exceptional event the specified interval updates still terminate. The cost follows from Section 5.1. □

A Bernoulli coin of probability v is obtained by the same one-threshold construction, with error at most 2^{-t} and polynomial deterministic cost. For positional skipping we use $K(1-r_b, N)$, where N is the remaining number of positions. Its value N means that no further position is proposed. For a retained prime we use $K(p^{-s}, e_{\max})$, whose terminal value represents exponent overflow. These terminal outcomes are retained as outcomes of the routine, rather than resampled.

6 The algorithm and its complexity

We can now specify one actual trial. The following routines refer to Lemma 5.1 and its Bernoulli counterpart, each with error at most $\delta = 2^{-t}$, where

$$t = h + H_0 + 2B + 10, \qquad H_0 = 32768B^3. \tag{6.1}$$

Compute J once from (3.3). Select j exactly and uniformly from J, using rejection from a power-of-two range; this takes fewer than two rounds in expectation and $O(\log B)$ expected random bits. Start the trial with $s = j/B$, $n = 1$ and an empty factor list.

Algorithm 1: one finite-bit trial.

1. For $b = 1, \dots, \lfloor \log_2 y \rfloor$, set $\mathrm{pos} = 2^b$ and $\mathrm{hi} = \min(2^{b+1} - 1, y)$.
2. While $\mathrm{pos} \le \mathrm{hi}$, put $N = \mathrm{hi} - \mathrm{pos} + 1$ and draw $K(1 - 2^{-bs}, N)$ using the δ-accurate routine. If its value is N, finish this block. Otherwise let $m = \mathrm{pos} + K$ and set $\mathrm{pos} = m + 1$.
3. Toss the δ-accurate coin of probability $(2^b/m)^s$. If it fails, continue the block. If it succeeds, apply the exact primality test to m; continue the block if m is composite.
4. For prime m, compute $e_{\max} = \max\{e \ge 0 : nm^e \le X\}$. If $e_{\max} = 0$, reject the trial. Otherwise draw $K(m^{-s}, e_{\max})$ with the δ-accurate routine. If its value is $e_{\max}$, reject the trial. Otherwise set $e = 1 + K$, replace n by nm^e, append (m, e) to the factor list, and continue the block.
5. After all blocks have finished, toss the δ-accurate coin of probability $(n/X)^s$. On success return n and its factor list; on failure reject the trial.

Repeat independent trials, with a new independent uniform choice from J each time, until one is accepted. In steps 3 and 4, continuing the block returns to step 2; finishing a block advances the outer loop in step 1.

Proof of Theorem 1.1. The only approximated decisions in a trial are positional skips, thinning coins, exponent draws and the final coin. Each integer position is proposed at most once, whatever numerical approximations occur. There are at most three primitive calls per proposed position, one terminal skip per block and one final coin. Their number Q therefore has the deterministic bound

$$Q \le 3X + B + 1 \le 5X. \tag{6.2}$$

The exact parameter-selection loop is not included in this bound; its expected cost was accounted for separately and it can be coupled identically in the ideal and numerical algorithms.

Couple each approximate primitive maximally to its ideal counterpart until the first disagreement. Before that disagreement their inputs coincide, including inputs chosen adaptively from earlier outputs. The probability η of any disagreement during the trial is at most

$$\eta \le 5X\delta \le \frac{5}{1024}\, 2^{-h} 2^{-H_0}/X, \tag{6.3}$$

where $X \le 2^B$ was used. If $\widetilde{a}$ denotes the numerical acceptance probability and a its ideal value, then $|\widetilde{a} - a| \le \eta$. Lemma 4.2 implies $\eta < a/2$, hence $\widetilde{a} \ge a/2$.

For completeness, if two joint laws of an output and an acceptance flag differ by at most η in total variation and the first acceptance probability is $a > 0$, their conditional output laws differ by at most $2\eta/a$. For an event E of accepted outputs, write the two subprobabilities as $p(E), \widetilde{p}(E)$ and their masses as $a, \widetilde{a}$. Then

$$\left| \frac{p(E)}{a} - \frac{\widetilde{p}(E)}{\widetilde{a}} \right| \le \frac{|p(E) - \widetilde{p}(E)|}{a} + \frac{\widetilde{p}(E)|a - \widetilde{a}|}{a\widetilde{a}} \le \frac{2\eta}{a}. \tag{6.4}$$

The ideal conditional law is $U_{X,y}$ by (2.2). Combining (6.3), (6.4) and $a \ge 2^{-H_0}$ gives the error bound (1.1). Independent repetition until acceptance produces precisely this numerical conditional law; it does not add a fresh error for every failed trial.

It remains to control expected work, since the worst-case bound (6.2) is exponential in B. For the ideal trial, (3.4) gives

$$\mathbb{E}Q_{\mathrm{ideal}} \le 3(2048B^2) + B + 1 \le 6146B^2.$$

Under the same transcript coupling, the two call counts agree whenever no disagreement occurs and differ by at most $5X$ otherwise. Consequently

$$\mathbb{E}Q_{\text{actual}} \le 6146B^2 + 5X\eta \le 6146B^2 + 1. \tag{6.5}$$

The last inequality follows directly from (6.1) and $X \le 2^B$: $5X\eta \le 25X^2 2^{-h-H_0-2B-10} < 1$. Each primitive, primality test and intervening arithmetic operation costs a fixed polynomial in $B + t$, and $t = O(B^3 + h)$. Thus the expected cost of a single numerical trial is polynomial in $B + h$.

By Proposition 4.1, $\widetilde{a} \ge c_0/(2B^2)$. If C is the cost of one trial, the expected cost of independent repetition is $\mathbb{E}C/\widetilde{a}$. To see this without assuming that a trial's cost and success are independent, sum the costs of trials reached: the event of reaching trial i depends only on earlier trials and is independent of trial i's cost. The geometric sum of those probabilities is $1/\widetilde{a}$. This proves the asserted polynomial expected bit bound. Positive acceptance probability also gives almost-sure termination.

Finally, numerical probability errors cannot create an invalid output. The primality test, cutoff, multiplication and overflow checks are exact. Proposed bases increase through disjoint blocks, so each prime is recorded at most once, and the empty list is the correct factorization of 1. The factorization has polynomial encoded length because $\sum_p e_p \log p \le \log X$. The theorem follows. □

7 Scope of the result

The construction establishes nearly uniform factored sampling with one expected polynomial bit bound across all smoothness cutoffs. The parameter grid avoids computing the saddle point, while the sparse proposals avoid processing a supplied list of every eligible prime. The separate precision argument ensures that these ideal sampling observations survive in a fair-bit model.

The constants used for precision are deliberately conservative. Neither a competitive implementation nor a practical running-time improvement over the algorithms in [1–3] follows from the proof. The integer and its factorization are exact, but the distribution has the requested total-variation tolerance; exact uniform generation from fair bits is not asserted here.

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No datasets were generated or analyzed in this theoretical study.

Declaration of generative AI and AI-assisted technologies

AI tools assisted the formulation, literature comparison, proof development, and preparation of this manuscript within the Mathematics Knowledge and Connections Program. AI tools were used to propose and examine arguments, compare the result with published sources, check proof steps, and revise exposition. The author directed the research selection and manuscript revisions and is responsible for the content and its final verification.

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